

PYTHON FOR BEGINNERS AND PROFESSIONALS

SESSION 03

Conditional Statements with Practice Questions

What is a Conditional Statement?

In real life, we make decisions based on conditions – *"If it is raining, carry an umbrella, otherwise don't."* Python works the same way. A conditional statement lets your program make decisions and execute different code based on whether a condition is **True** or **False**.

```
age = 20

# Program decides what to print based on the value of age
if age >= 18:
    print("You are eligible to vote.")
```

IF Statement

The *if* statement is the most basic form. If the condition is *True*, the indented block runs. If it is *False*, Python simply skips it.

```
marks = 75

if marks >= 40:
    print("You have passed.")

# If marks were 30, nothing would print at all
```

Watch Out: Indentation is not optional in Python. The block inside *if* must be indented usually 4 spaces. Missing indentation will give you an *IndentationError*.

IF-ELSE Statement

Use **else** when you want something to happen when the condition is **False**.
One of the two blocks will always run – never both, never neither.

```
age = 16

if age >= 18:
    print("You can vote.")
else:
    print("You are too young to vote.")
```

IF-ELIF-ELSE Statement

When you have **more than two** possible outcomes, use elif (short for "else if"). Python checks each condition from top to bottom and runs the first one that is True. The rest are skipped entirely.

```
marks = 82

if marks >= 90:
    print("Grade: A")
elif marks >= 75:
    print("Grade: B")
elif marks >= 60:
    print("Grade: C")
elif marks >= 40:
    print("Grade: D")
else:
    print("Grade: F")

# Output: Grade: B
```

NESTED IF-ELSE Statement

You can place an *if* statement inside another *if* statement. This is called nesting and is useful when a second condition only makes sense if the first one is already True.

```
age          = 22
has_degree   = True

if age >= 18:
    print("Age requirement met.")
    if has_degree:
        print("You are eligible for this job.")
    else:
        print("You need a degree for this job.")
else:
    print("You are too young to apply.")
```

Shorthand if-else (Ternary Operator)

Python lets you write a simple if-else in a single line. This is called the ternary operator. It is useful when you want to assign a value based on a condition.

```
# Normal way
if age >= 18:
    status = "Adult"
else:
    status = "Minor"

# Shorthand way (same result, one line)
status = "Adult" if age >= 18 else "Minor"
print(status)
```

Practice Questions

Q7: Take a number as input. Print whether it is positive, negative, or zero.

Q8: Take two numbers as input. Print the greater of the two. If they are equal, print "Both are equal."

Q9: Take a student's marks as input. Print their grade based on this scale:

- 90 and above → A
- 75 to 89 → B
- 60 to 74 → C
- 40 to 59 → D
- Below 40 → F

Q10: Take a year as input. Check if it is a leap year. A year is a leap year if it is divisible by 4, but not by 100, unless it is also divisible by 400.

Practice Questions (Homework)

Q11: Take a person's age and whether they have a valid ID (True/False) as input. They can enter a venue only if they are 18 or older AND have a valid ID. Print the appropriate message.

Q12: Take three numbers as input. Print the largest of the three without using any built-in function.

Q13: Take a number as input. Using the ternary operator, print "Even" or "Odd" in a single line.

Q14: A shop gives discounts based on purchase amount:

- Above 5000 → 20% discount
- Above 2000 → 10% discount
- Above 1000 → 5% discount
- 1000 or below → no discount